PROJECT SYN // ZERO-TRUST

Hardware-Secured AI & Telemetry Integrity on IBM LinuxONE

CPACF WireGate, Chained SQLite WAL Ledger & Adversary Spoof Neutralization

Track: AI Secured: Innovation Without Exposure | IBM Z Datathon 2026 Championship Deck

// ZERO-TRUST THREAT VECTOR // IBM Z DATATHON 2026

Threat Model: Cyber-Physical Attacks on Critical Healthcare

Protecting AI Decision Engines from Injected Falsehoods

Telemetry Spoofing

RISK: CRITICAL

- Adversaries forge ICU bed availability & oxygen reserve metrics.
- Objective: Trigger artificial hospital saturation and divert critical care.
- Vulnerability: Standard cloud APIs ingest unauthenticated JSON.

Replay & Delay Injections

RISK: HIGH

- Attackers capture valid historical packets and replay during flood surges.
- Masquerades dead substations as fully energized to delay utility rescue.
- Defeats simple static API token authentication.

AI Model Poisoning

RISK: FATAL

- Injecting manipulated flood levels biases machine learning models.
- Causes ambulance routing algorithms to misdirect trauma convoys.
- Result: Catastrophic physical consequences in disaster zones.

// CRYPTOGRAPHIC DEFENSE // IBM Z DATATHON 2026

CPACF-WireGate: Wire-Speed Hardware Verification

Native IBM z16 Hardware Instruction Acceleration (<0.24ms)

Zero-Trust Ingestion Engine

WIRE SPEED: < 0.24 ms

- Every inbound IoT telemetry packet is authenticated BEFORE reaching AI.
- Native IBM CPACF KMAC acceleration computes HMAC-SHA256 in <0.24ms.
- Monotonic nonce cache rejects replayed and delayed packets.
- Tampered signatures immediately trigger 401 Unauthorized rejection.
- Zero CPU overhead compared to standard software cryptographic libraries.

Chained Forensic WAL Ledger

CHAIN CAPACITY: 1,200+ req/s

- Ingestion queue buffers over 1,200 requests/second with zero packet drop.
- SQLite Write-Ahead Logging (WAL) with cryptographic hash chaining:
- Hash_n = SHA256(Hash_{n-1} + Nonce + TelemetryPayload).
- Provides an immutable, tamper-evident audit trail for disaster forensics.
- P2P cross-node pinning between Node 1 (KEM) and Node 2 (Tata Memorial).

Live Demonstration: Simulated Cyber Attack Defeated

Interactive Red-Button Attack Trigger on Live Mainframe Dashboard

1. Operator Injects Spoof

STEP 1: INJECTION

- Dispatcher clicks red button: [SIMULATE TELEMETRY INJECTION].
- Injects tampered grid voltage & falsified ICU bed metric.
- Payload carries altered HMAC signature.

2. CPACF Hardware Drop

STEP 2: 180us DROP

- Mainframe CPACF gate detects cryptographic signature mismatch.
- Packet dropped in 180 microseconds without touching database.
- HTTP 401 Unauthorized returned instantly.

3. UI Breach Notification

STEP 3: LEDGER PINNED

- UI flashes: [HMAC AUTHENTICATION FAILED - INJECTION DROPPED].
- Forensic breach incident logged to immutable chained ledger.
- Zero corruption of clinical triage or ambulance routing engines.

Production Mainframe Architecture & Verifiable Links

Full Alignment with IBM Z Datathon Guidebook & Rubric

Bare-Metal s390x Infrastructure

DEPLOYED ON s390x SILICON

- Host: IBM LinuxONE III (Machine 9175 z16 Architecture).
- Node 1: KEM Hospital Apex -> http://148.100.79.88:8000/ui/
- Node 2: Tata Memorial Hub -> http://148.100.98.9:8000/ui/
- Public System Guide: https://yuvrajlokhande19.github.io/Project-Syn-IBM-LinuxONE/
- OpenAPI Swagger Documentation: http://148.100.79.88:8000/docs
- GitHub Clean Repo: https://github.com/yuvrajlokhande19/Project-Syn-IBM-LinuxONE.git

5-Part JupyterLab Verification Suite

REPRODUCIBLE BENCHMARKS

- Live on Node 1: http://148.100.79.88:38888/?token=Your_Token1
- 00_START_HERE: s390x CPU Info, Big-Endian & CPACF Flags.
- 01_OpenData: OpenCity CKAN Real Mumbai Hospital API Harvesting.
- 02_SnapML: <0.8ms Random Forest Anomaly Detection Benchmarks.
- 03_Dijkstra: Deterministic Dynamic Elevated Routing Validation.
- 04_CPACF: 10,000-Packet WireGate Spoof Rejection Benchmark.